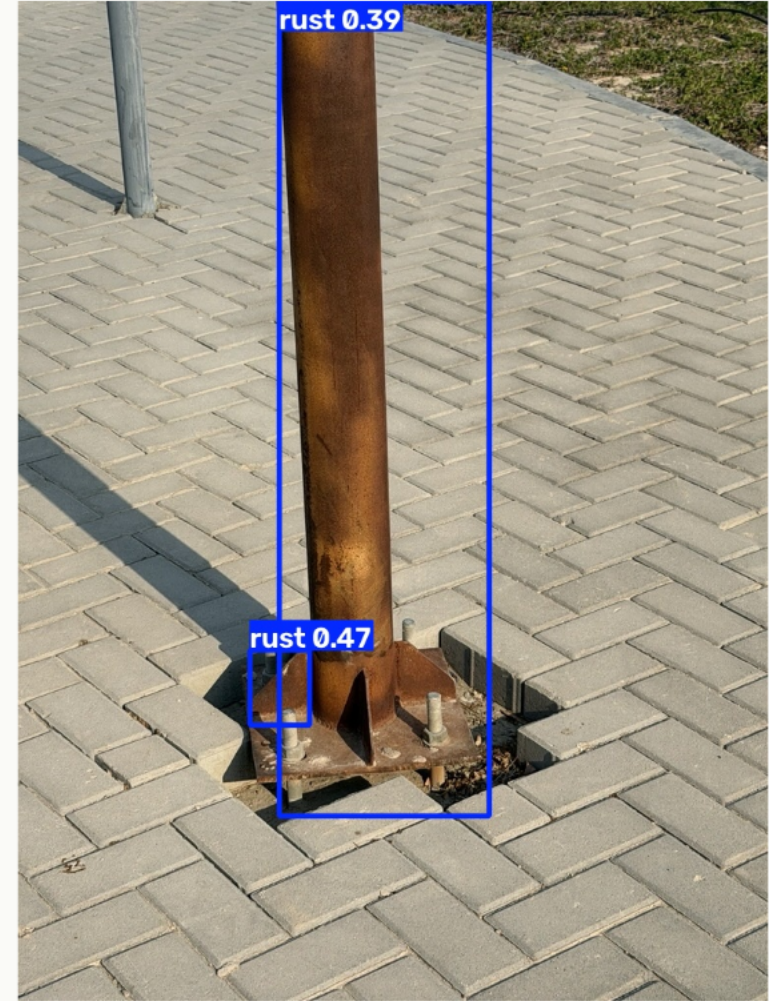


Steel rust detection

YOLOv8 inspection prototype

Mohammad Mango
Module 4, Unit 3
Computer Vision
24 September 2026

Candidate rust regions for human review



Dataset and labeling policy

440 images

4,107 stored boxes

One class: rust

352 training / 88 validation

Seed 42 fixes the split.

No independent test set.

Tight boxes around visible rust.

Exclude paint, dirt and concrete stains.

Inherited broad boxes remain a limitation.



Source annotation example

Reproduction and training

Public Release ZIP + SHA256
Version 1, fixed YOLOv8 export
No dataset API key or Drive mount

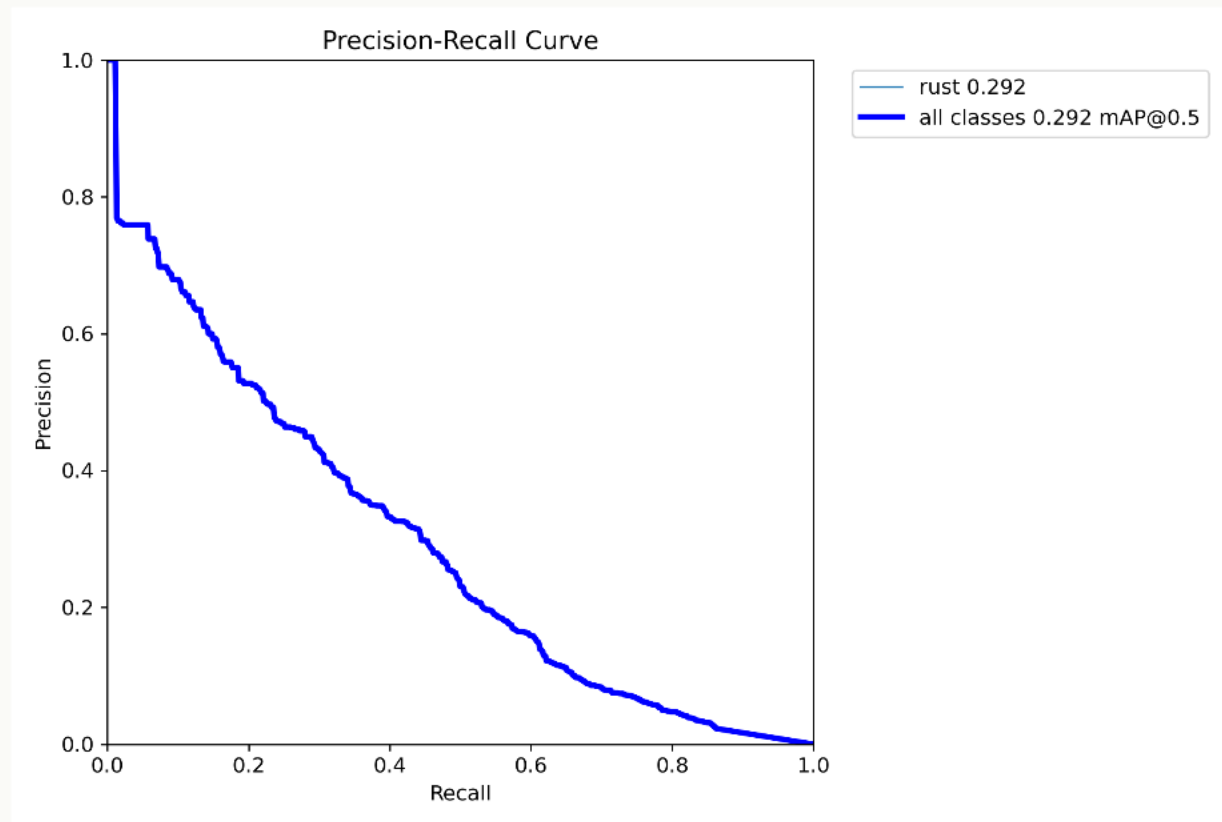
YOLOv8n, 30 epochs
512 px, batch 16, seed 42
Ultralytics 8.3.221, Tesla T4

Training run: 23 September 2026
207.98 seconds for the training call
Author confirmed a fresh GitHub-to-Colab run

CPU inference also completed on 24 September.
45.45 seconds after setup, using the published best.pt.
Inference reuses trained weights and performs zero training epochs.

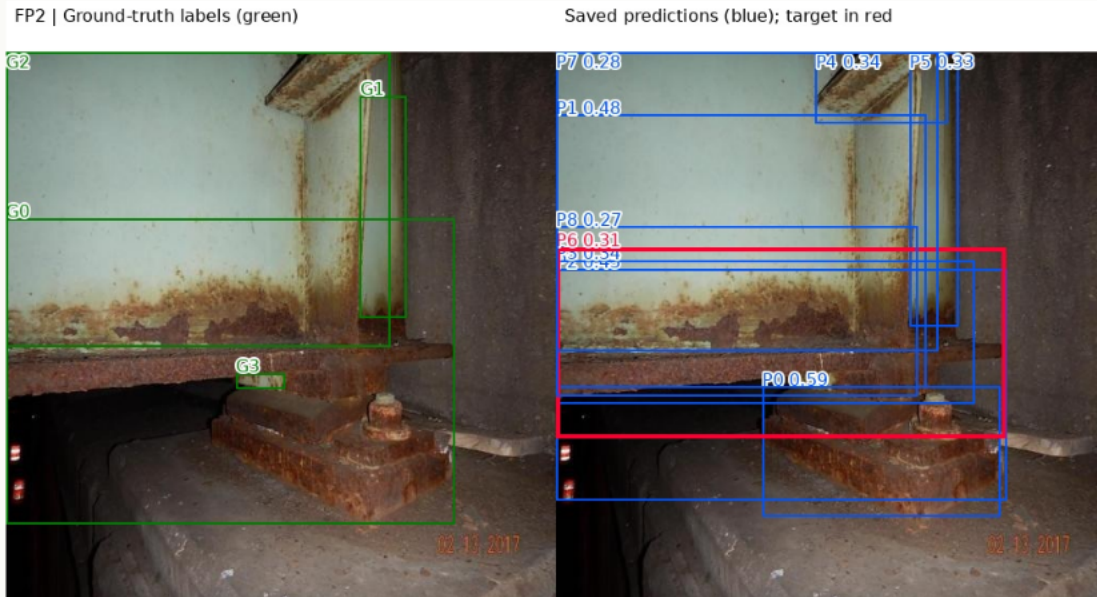
Validation performance

Metric	Result
Precision	35.51%
Recall	36.49%
mAP50	29.23%
mAP50-95	12.24%



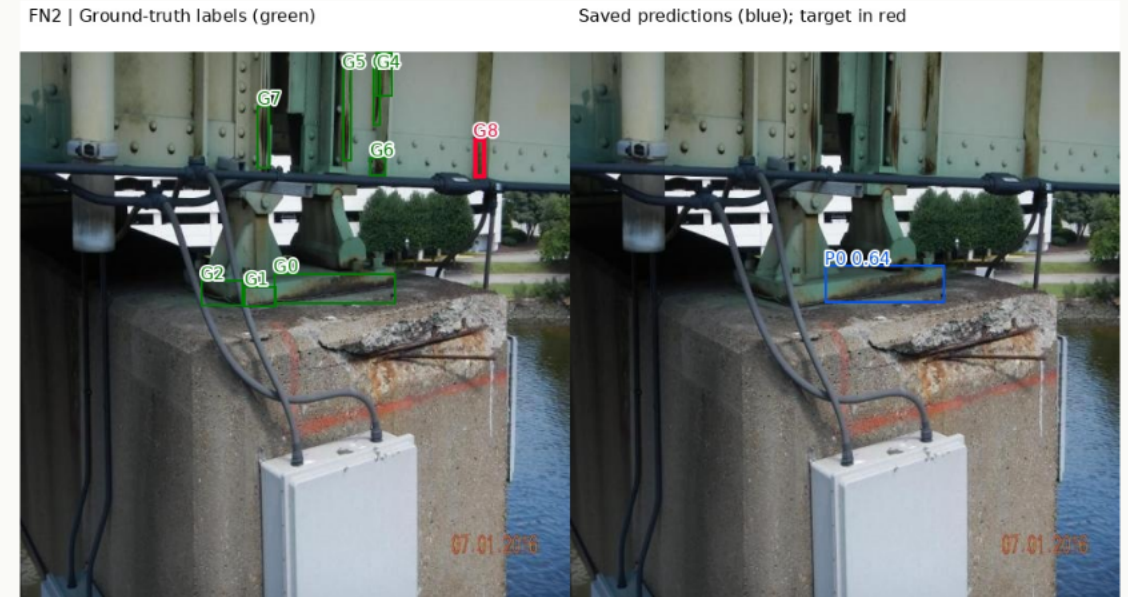
88 images / 707 instances. Stricter overlap requirements reduce mAP.
Substantial missed regions and false detections limit practical use.

Error analysis



Duplicate detection

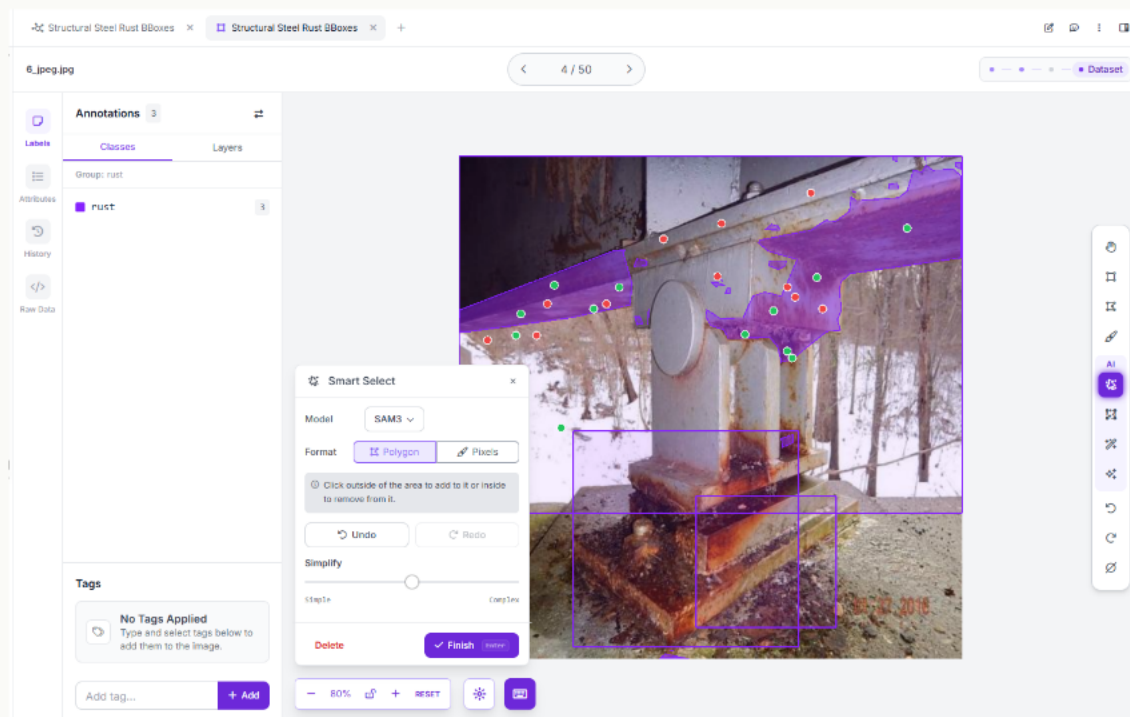
An extra box matches a region already assigned to another prediction.



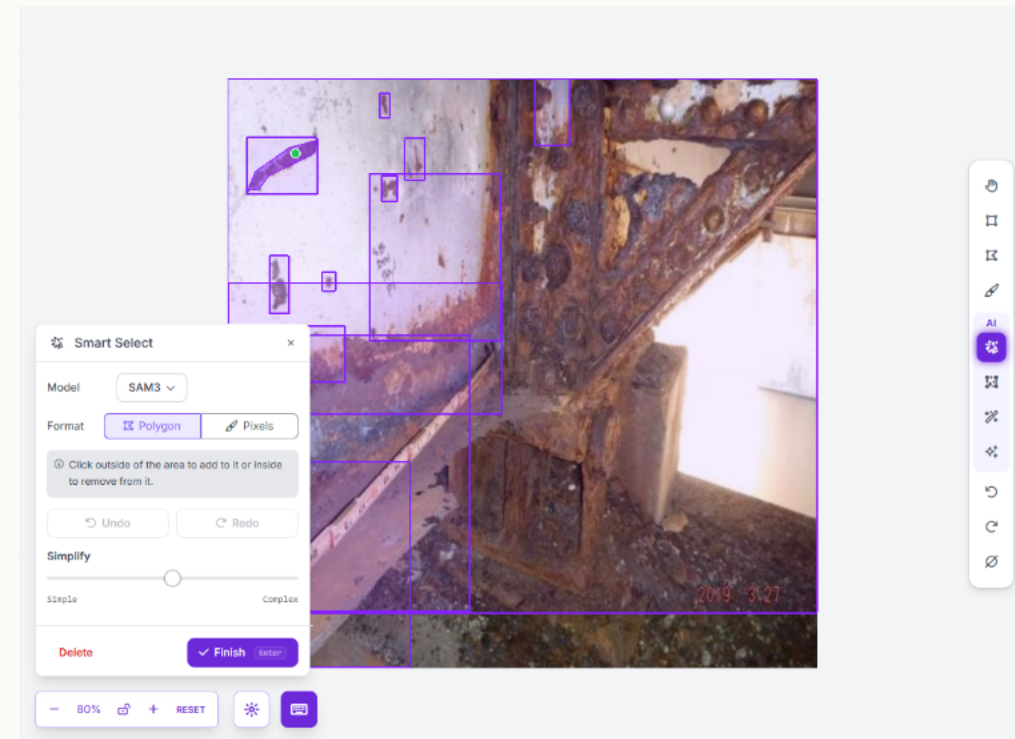
Missed thin rust region

The model misses the narrow upper region at confidence 0.25.

SAM3 annotation exploration



Complex corrosion: repeated corrections still leave imperfect boundaries.



Isolated patch: one point produces a visually close starting polygon.

Two qualitative trials. No mask accuracy score or retraining claim.

New photographs and responsible use



Example with no detections at confidence 0.25

Five author photographs from public places
provide qualitative checks.

No independent ground-truth labels.
No detections cannot establish healthy steel.
Photos cannot establish corrosion depth
or structural safety.

Human inspection remains essential.

Priorities and conclusion

1 Audit similar images and split by structure/site

Reduce leakage risk before measuring generalization to new sites.

2 Standardize rust-patch annotations

Resolve duplicate boxes and broad component-sized labels.

3 Add difficult examples and verified negatives

Target small rust, shadows, red paint and healthy or dirty steel.

Academic baseline completed. Further data work is proposed.